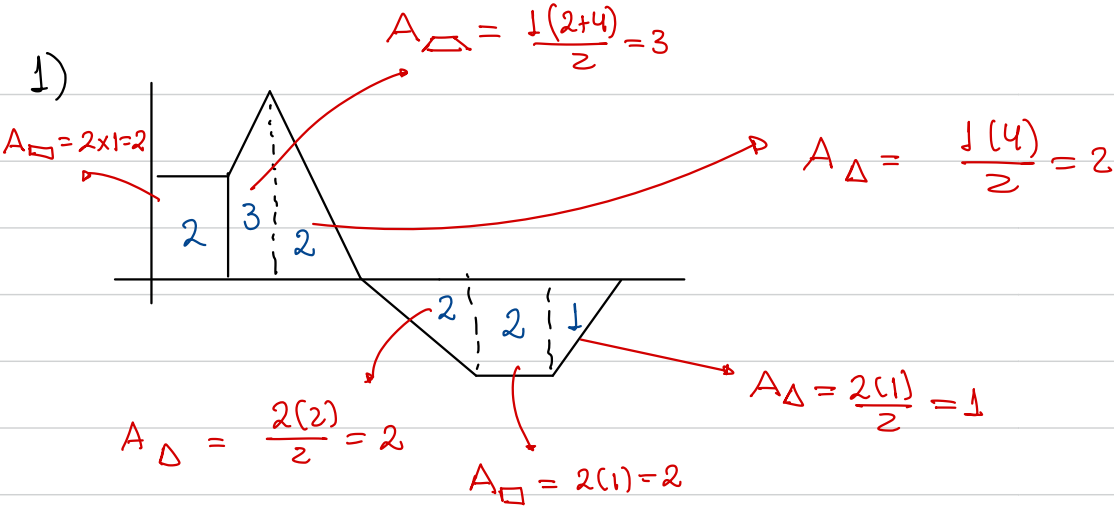
1)



daí $\int_0^7 f\,dx = \cancel{2} + 3 + \cancel{2} - \cancel{2} - \cancel{2} - 1 = 2.$

2) $$\frac{d}{dx}\int_{2x+1}^{3x^2} \operatorname{sen} u \cos u \, du = \operatorname{sen}(3x^2)\cos(3x^2)\frac{d}{dx}(3x^2) - \sin(2x+1)\cos(2x+1)\frac{d}{dx}(2x+1)$$

$$= 6x\sin(3x^2)\cos(3x^2) - 2\operatorname{sen}(2x+1)\cos(2x+1)$$

3) $$\int \sin^2(2x)\,dx = \int \frac{1-\cos(4x)}{2}\,dx = \frac{x}{2} - \frac{\operatorname{sen}(4x)}{8} + c$$

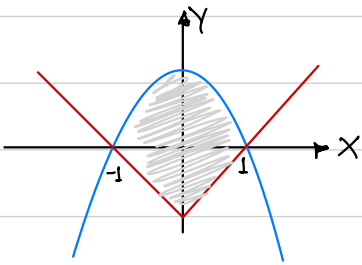
4) $\int_0^{\pi/2} \operatorname{sen}^3 x \cos x \, dx$ fazer $u = \operatorname{sen} x \Rightarrow du = \cos x\, dx$

$x = 0 \Rightarrow u = 0$

$x = \pi/2 \Rightarrow u = 1$

daí $\int_0^{\pi/2} \operatorname{sen}^3 x \cos x\, dx = \int_0^1 u^3\, du = \frac{u^4}{4}\Big|_0^1 = \frac{1}{4}$

5)



$1 - x^2 = |x| - 1$

Se $x \geq 0 \Rightarrow |x| - 1 = x - 1$

daí $1 - x^2 = x - 1 \Rightarrow x^2 + x - 2 = 0$

$\Rightarrow (x+2)(x-1) = 0$

$\Rightarrow \underline{x = -2}$ ou $x = 1$

não é possível pois $x \geq 0$

daí $x = 1$.

Se $x < 0 \Rightarrow |x| - 1 = -x - 1$

daí $1 - x^2 = -x - 1$

$\Rightarrow x^2 - x - 2 = 0$

$\Rightarrow (x-2)(x+1) = 0$

$\Rightarrow \underline{x = 2}$ ou $x = -1$

não é possível pois $x < 0$

daí $x = -1$.

logo temos $x = -1$ e $x = 1$

daí $A = \int_{-1}^{1} 1 - x^2 - (|x| - 1)\, dx = 2\int_0^1 1 - x^2 - x + 1\, dx$

$= 2\int_0^1 2 - x - x^2\, dx = 2\left(2x - \frac{x^2}{2} - \frac{x^3}{3} \Big|_0^1 \right)$

$= 2\left(2 - 1/2 - 1/3 \right) = \cancel{2}^{1}\left(\frac{12 - 3 - 2}{\cancel{6}\,3} \right) = \underbrace{7/3}_{\text{é o valor da área.}}$